FolliCare AI: Architecture & Component Specification

End-to-End Technical Documentation: Frontend CopilotKit, LangGraph Multi-Agent System, BGE-M3 Embeddings, Voice STT/TTS, MCP Server & Observability Pipeline

Executive Summary: *FolliCare AI is an enterprise on-device AI commerce and clinical trichology platform. It couples a React 18 storefront with CopilotKit state synchronization, a LangGraph hierarchical multi-agent core, local Ollama LLMs, BGE-M3 (1024-dim) vector embeddings, browser-native voice STT/TTS, and a JSON-RPC 2.0 Model Context Protocol (MCP) server with 100% cloud-independent local execution ($0 API cost).*

1. End-to-End Technology Stack Matrix

Layer	Core Technologies	Version / Specs	Role & Function in FolliCare AI
Frontend Framework	React 18, Vite, Vanilla CSS Modules, Lucide React	React 18.2, Vite 5.x	High-performance storefront, CSS custom properties dynamic theming, interactive modals & drawers.
Frontend AI Bridge	CopilotKit Protocol (useCopilotReadable, useCopilotAction)	Custom zero-overhead	Bidirectional state sync between React DOM/context and the LangGraph multi-agent core.
Voice Processing	W3C Web Speech API (SpeechRecognition, SpeechSynthesis)	Native Browser API	Zero-latency STT voice input with mic pulse and natural speech synthesis audio playback.
Multimodal Vision	Qwen 2.5-VL (7B Vision-Language), Base64 Image Ingestion	Ollama qwen2.5vl:7b	Direct visual diagnostics of scalp density, sebum buildup, follicle miniaturization, and texture.
Agent Orchestration	LangGraph (StateGraph, START, END), Supervisor Router	LangGraph 0.2+, Py 3.10+	Hierarchical state management, intent routing, and self-correction reflection loops.
Local LLM Runtime	Ollama Local LLM Engine (HTTP 127.0.0.1:11434)	qwen2.5:3b, qwen2.5-coder:7b	qwen2.5:3b (Supervisor/RAG), qwen2.5-coder:7b (React AST Developer Mode), qwen2.5vl:7b (Vision).
Vector Embeddings	BAAI BGE-M3 (BAAI/bge-m3)	1024-dim dense vectors	Multi-lingual, multi-granularity (8192 token window) dense embeddings of clinical & catalog data.
Vector Storage	ChromaDB Persistent Client	HNSW Cosine Index	Persistent vector store with multi-field hard metadata filtering ($and clauses).
Semantic Reranker	Cross-Encoder Sentence Transformers	Joint Cross-Attention	Stage-2 precision reranking of top candidate chunks retrieved from ChromaDB.
Tool Calling Protocol	Model Context Protocol (MCP)	JSON-RPC 2.0 Standard	Standardized interface exposing 4 tools and 2 live data resources for external & local AI agents.
Backend Gateway	FastAPI, Uvicorn, Pydantic v2	Async ASGI (:8000)	High-concurrency REST & CopilotKit gateway (/api/chat, /api/products, /copilotkit).
Relational DB	SQLite3 with Write-Ahead Logging (WAL)	Native Python sqlite3	Persistent catalog storage, evaluations, telemetry traces, tool call logs, and customer hindsight memory.
Observability & Eval	OpenTelemetry Tracing Pipeline, Streamlit	Port 8502 Dashboard	Real-time tracking of token throughput, latency percentiles, tool precision, and RAG Triad scores.

2. High-Level Architectural Flow & Request Lifecycle

The system operates on an event-driven bidirectional flow between the user interface and local agent nodes:

- **1. Multimodal Input Capture:** The user interacts via voice mic (Web Speech STT), camera upload (Base64 JPEG), or text chat.
- **2. CopilotKit State Serialization:** React collects active DOM context via useCopilotReadable and posts to /api/chat.
- **3. Supervisor Intent Classification:** LangGraph supervisor routes query into DEVELOPER_UI_CODE, PRODUCT_INGESTION, UI_MUTATION, KNOWLEDGE_QUERY, ROUTINE_CONSULT, or HYBRID.
- **4. Two-Stage RAG Execution:** 1024-dim BGE-M3 query vector searches ChromaDB with metadata filters, followed by Cross-Encoder reranking.
- **5. UI Action Synthesis & Verification:** Generates structured ui_commands verified by the Tool Validator reflection loop.
- **6. Response Synthesis & Speech Dispatch:** Synthesizes Markdown response + clean speech text for browser-native Web Speech TTS (■).
- **7. State Mutation & Hindsight Logging:** Frontend mutates theme/cart; telemetry is recorded in SQLite for LLM-as-a-Judge evaluation.

3. Low-Level Component-by-Component Deep Dive

Component 1: Multi-Agent Supervisor & LangGraph Engine

Source File(s): agents/orchestrator.py, agents/langgraph_workflow.py

- Implements a formal LangGraph StateGraph parameterized by FolliCareState.
- Includes conditional edge routing and an automated self-correcting reflection loop (up to 3 retries).
- Manages session tracking, user hindsight memory injection, and execution path telemetry.

Component 2: Two-Stage Hybrid Vector Store & Semantic Chunking

Source File(s): rag/chroma_vector_store.py, rag/chunker.py, rag/reranker.py

- **Semantic Field Chunking:** Splits products into distinct {id}_overview and {id}_usage chunks to isolate active ingredients, pricing, and clinical studies.
- **BGE-M3 Dense Embeddings:** Generates 1024-dimensional normalized dense vectors with 8,192 token context support.
- **Multi-Field Metadata Filtering:** Executes hard $and constraints on category, concern, doc_type, and maximum price.
- **Cross-Encoder Reranker:** Performs joint query-document cross-attention scoring on top-8 candidates to extract top-3 precise snippets.

Component 3: Specialized Agent Suite

Source File(s): agents/rag_agent.py, agents/ui_agent.py, agents/react_coder_agent.py

- **RAG Trichology Agent:** Queries ChromaDB and formats clinical research citations.
- **UI Automation Agent:** Emits typed CopilotKit actions (theme switches, card scaling, cart additions, price overrides).
- **React Developer Agent:** Uses qwen2.5-coder:7b for JSX code generation, AST parsing, and theme token enforcement.
- **Product Ingestion Agent:** Ingests new formulas into SQLite and indexes vectors in real-time.
- **Response Synthesis Agent:** Formats markdown customer responses and cleans text for speech synthesis.

Component 4: CopilotKit React 18 Frontend & State Bridge

Source File(s): react_app/src/components/CopilotKitIntegration.jsx, react_app/src/App.jsx

- **useCopilotReadable:** Exposes live component state (active products, card dimensions, cart line items) to the AI.
- **useCopilotAction:** Exposes executable handlers (resizeCard, updatePrice, setTheme, addToCart).
- **CSS Theme Cascade:** Controls dynamic switching across ruby-red, ocean-blue, rose-luxury, and botanical-emerald with 0.4s transitions.

Component 5: Browser-Native Speech Recognition & Voice Synthesis

Source File(s): react_app/src/components/CopilotKitIntegration.jsx

- **Speech-to-Text (STT):** Built with window.webkitSpeechRecognition for continuous voice input with pulsating visual waves.
- **Text-to-Speech (TTS):** Uses SpeechSynthesisUtterance with regex Markdown/emoji stripping and pitch/rate tuning (1.05x).

Component 6: Multimodal Scalp Vision Diagnosis Engine

Source File(s): app.py, react_app/src/components/CopilotKitIntegration.jsx

- Dispatches Base64 scalp/hair photos to /api/multimodal/analyze.
- Ollama qwen2.5vl:7b performs visual trichological analysis (density, sebum, flaking) and pairs diagnosis with botanical regimens.

Component 7: Model Context Protocol (MCP) JSON-RPC 2.0 Server

Source File(s): mcp_server.py

- Exposes 4 standardized MCP tools: search_hair_knowledge, recommend_hair_routine, query_product_inventory, git_push_react_code.
- Exposes 2 live MCP data resources: follicare://catalog and follicare://clinical_studies.

Component 8: Long-Term User & Hindsight Memory Engine

Source File(s): agents/memory.py

- Tracks customer preferences (e.g. vegan/sulfate-free, hair thinning history) across conversations in a compact JSON schema.
- Injects personalized context into agent prompts without bloating context token count.

Component 9: Observability, Tracing & LLM-as-a-Judge Evaluation

Source File(s): agents/monitoring.py, agents/evaluator.py, metrics_dashboard.py

- Measures RAG Triad scores (Context Relevance: 92.2%, Faithfulness: 96.6%, Answer Relevance: 92.1%).
- Records token throughput (Input: 131.7 tok/s, Output: 43.9 tok/s), tool accuracy (97.9%), and reflection loop recoveries.
- Renders real-time telemetry on Streamlit Dashboard at http://localhost:8502.

Component 10: Persistent SQLite Database & Schema Sync

Source File(s): data/database.py, data/products.json

- Stores product catalog, evaluation logs, tool audit trails, and user memory.
- Maintains automated bidirectional synchronization between SQLite and data/products.json.

4. Server Ecosystem & Port Mapping

Service	Port / URL	Startup Command	Purpose & Responsibilities
React 18 CopilotKit	http://127.0.0.1:5173	npm run dev (react_app/)	Customer storefront with CopilotKit assistant, voice STT/TTS, and vision camera upload.
FastAPI Gateway	http://127.0.0.1:8000	uvicorn app:app --port 8000	Async REST API, LangGraph agent coordinator, CopilotKit runtime bridge.
Observability Dashboard	http://localhost:8502	streamlit run metrics_dashboard.py	Real-time RAG Triad scores, token throughput graphs, and tool trace logs.
Streamlit Storefront	http://localhost:8501	streamlit run streamlit_app.py	Alternative standalone Streamlit storefront for admin and fallback testing.
Ollama Inference Engine	http://127.0.0.1:11434	ollama serve	Local LLMs (qwen2.5:3b, qwen2.5-coder:7b, qwen2.5vl:7b, bge-m3 dense embeddings).